

Working with Attributes and Classes in JavaScript

JavaScript allows you to dynamically control HTML element attributes and CSS classes, enabling powerful, interactive web experiences.

1. Setting and Getting Attributes

Use `setAttribute()` to add or update an attribute and `getAttribute()` to retrieve its value.

Example:

```

```

```
const img = document.getElementById("myImage");

// Set a new attribute
img.setAttribute("alt", "Company Logo");

// Get an attribute value
console.log(img.getAttribute("src")); // "logo.png"

// Update an existing attribute
img.setAttribute("src", "new-logo.png");
```

2. Removing Attributes

Use `removeAttribute()` to completely remove an attribute from an element.

Example:

```
const img = document.getElementById("myImage");

// Remove the alt attribute
img.removeAttribute("alt");
```

3. Adding Classes

Use `classList.add()` to add one or more CSS classes to an element.

Example:

```
<div id="box"></div>
```

```
const box = document.getElementById("box");

box.classList.add("active");
box.classList.add("highlight", "shadow"); // Multiple classes
```

4. Removing Classes

Use `classList.remove()` to remove one or more classes.

```
box.classList.remove("highlight");
```

5. Toggling Classes

Use `classList.toggle()` to add a class if it doesn't exist, or remove it if it does.

```
box.classList.toggle("hidden");
```

6. Checking for a Class

Use `classList.contains()` to check whether an element has a specific class.

```
if (box.classList.contains("active")) {  
    console.log("The box is active");  
}
```

Summary

Task	Method
Set attribute	<code>element.setAttribute()</code>
Get attribute	<code>element.getAttribute()</code>
Remove attribute	<code>element.removeAttribute()</code>
Add class	<code>element.classList.add()</code>
Remove class	<code>element.classList.remove()</code>
Toggle class	<code>element.classList.toggle()</code>
Check class existence	<code>element.classList.contains()</code>